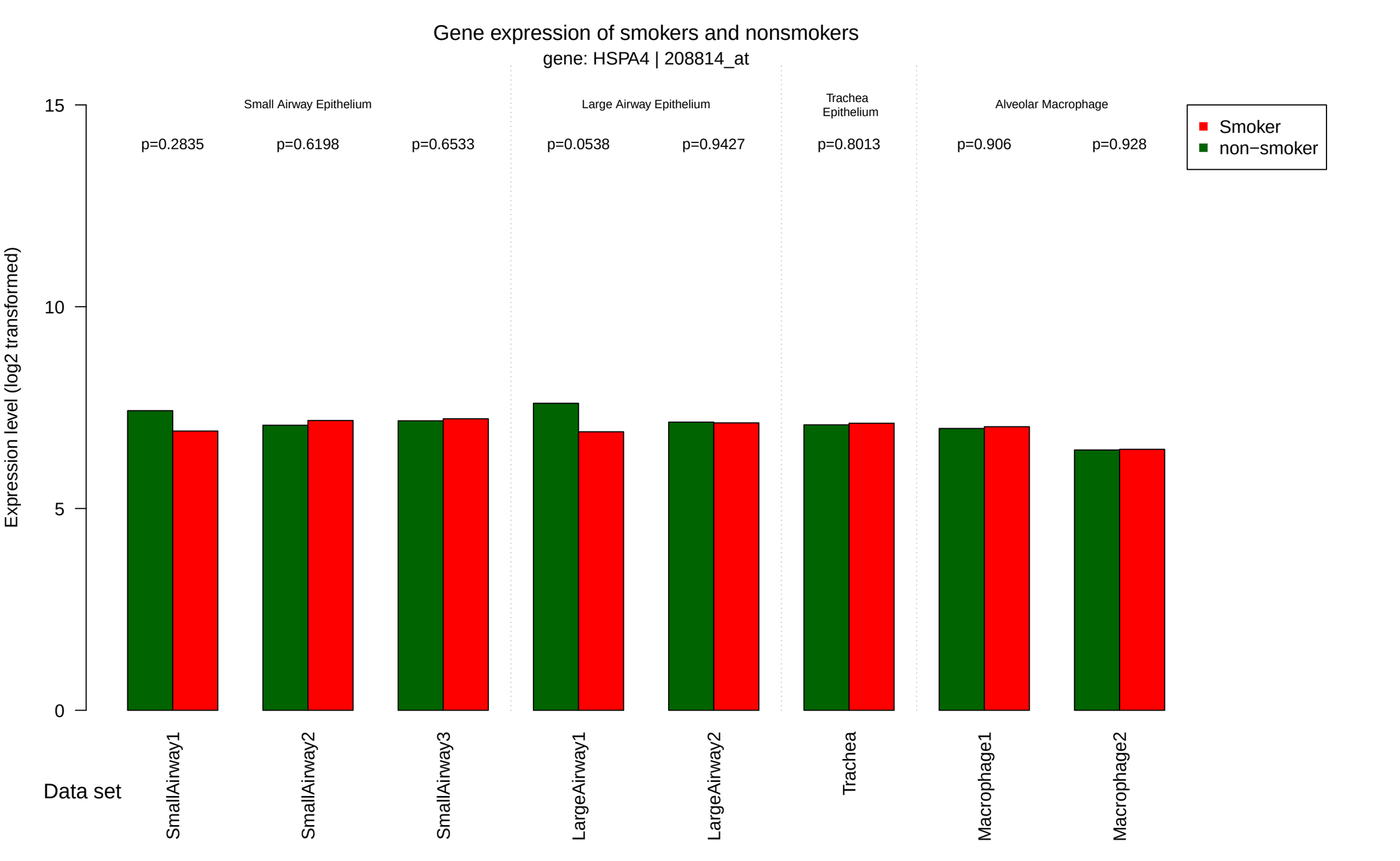
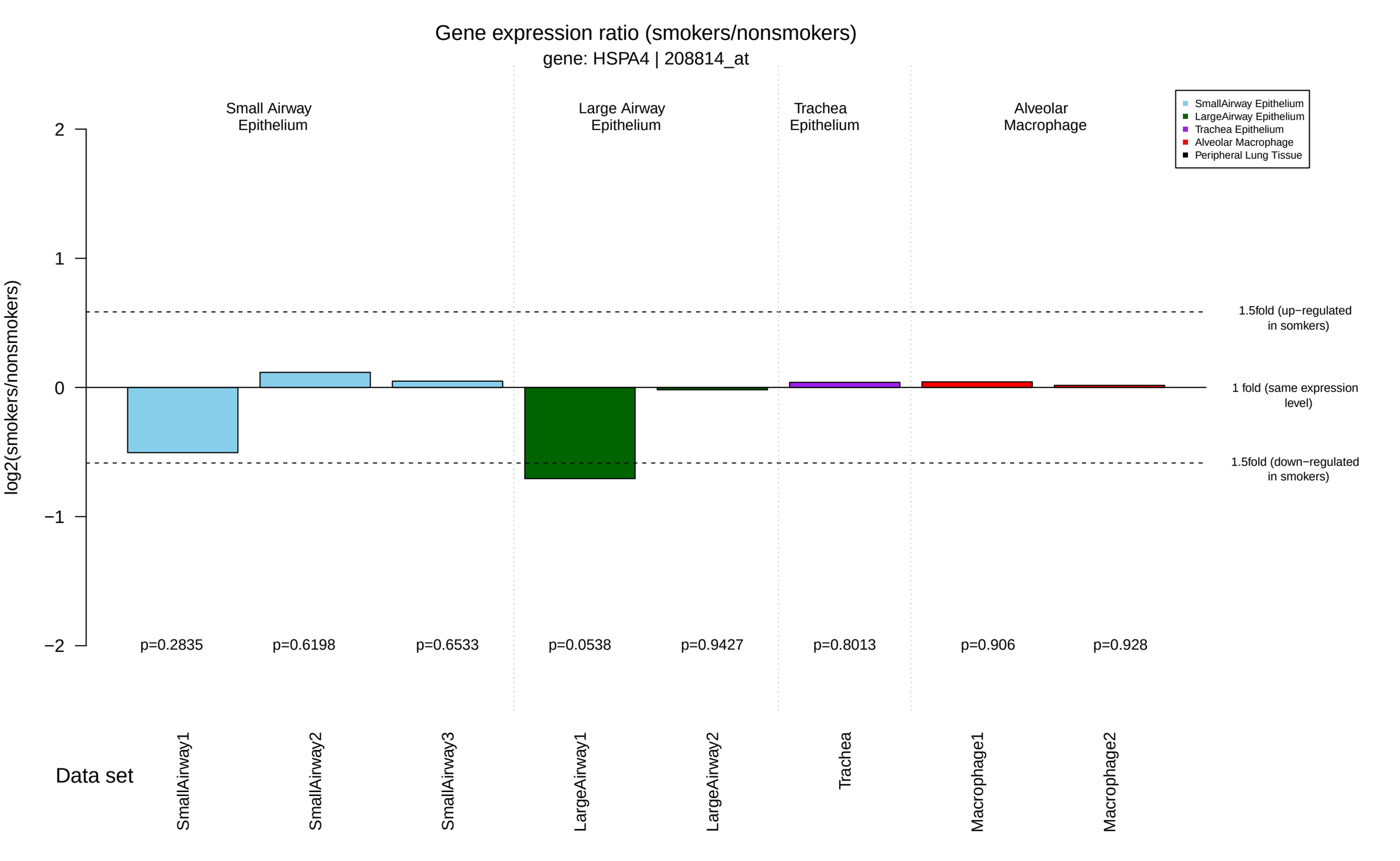
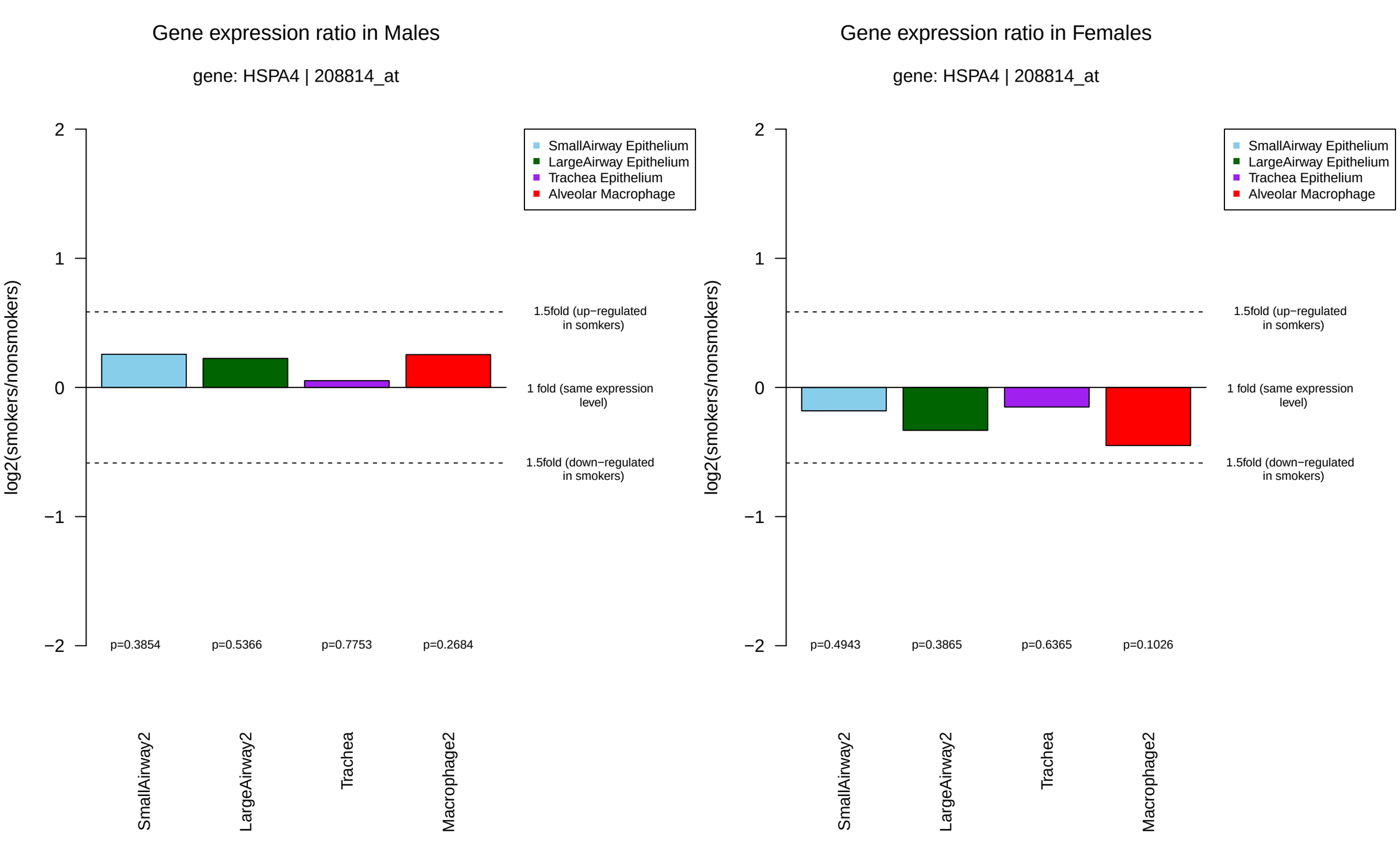
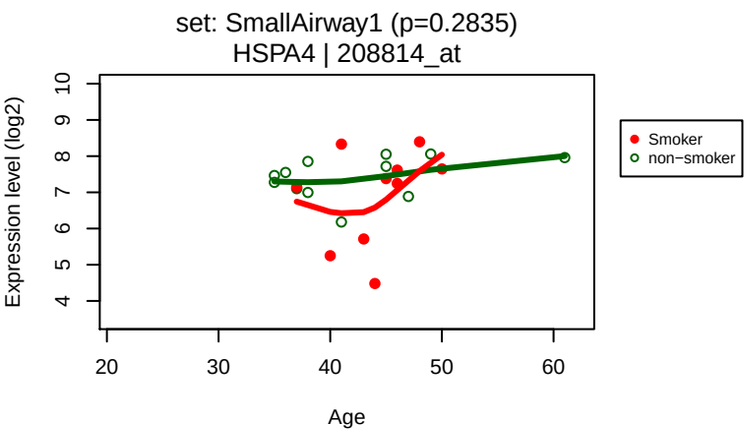
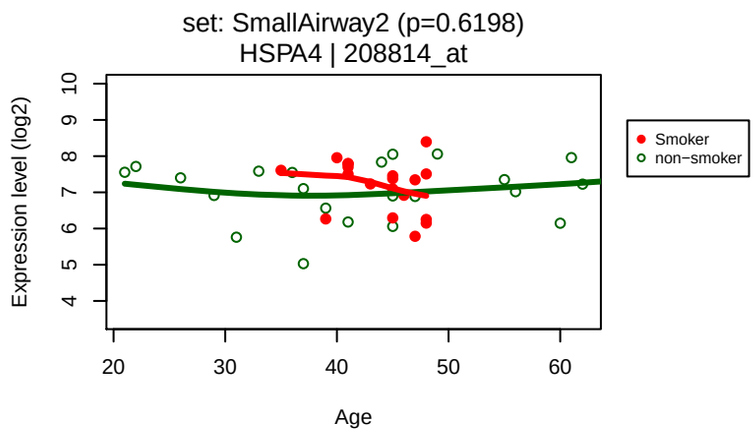
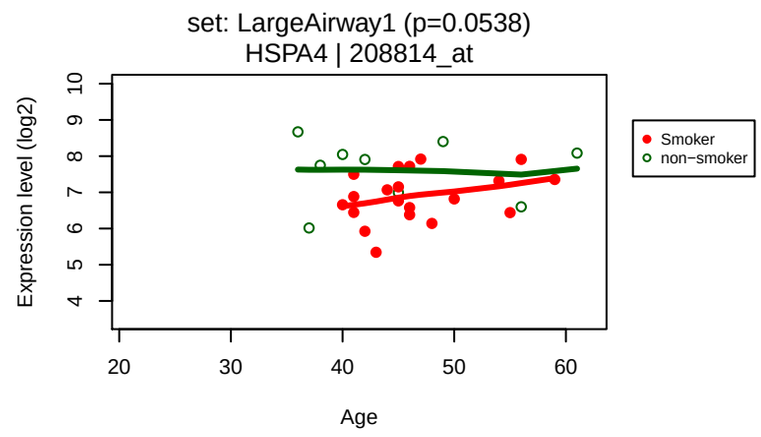

Gene expression of smokers and nonsmokers
gene: HSPA4 | 208814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2835
p=0.6198
p=0.6533
p=0.0538
p=0.9427
p=0.8013
p=0.906
p=0.928
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

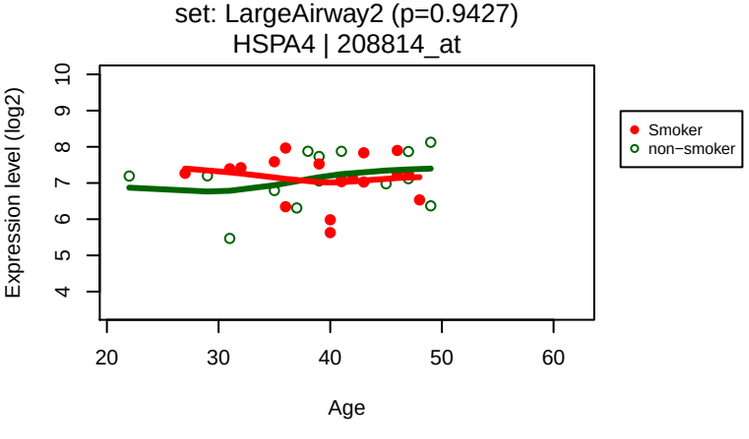

Gene expression ratio (smokers/nonsmokers)
gene: HSPA4 | 208814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2835
p=0.6198
p=0.6533
p=0.0538
p=0.9427
p=0.8013
p=0.906
p=0.928
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

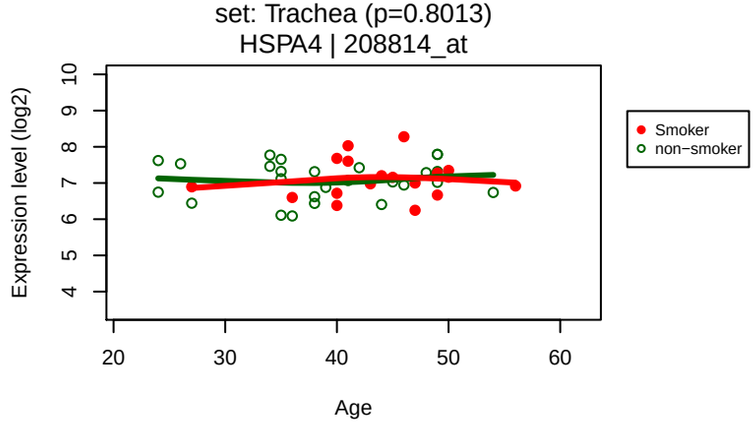

Gene expression ratio in Males
gene: HSPA4 | 208814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3854
p=0.5366
p=0.7753
p=0.2684
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA4 | 208814_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4943
p=0.3865
p=0.6365
p=0.1026
SmallAirway2
LargeAirway2
Trachea
Macrophage2

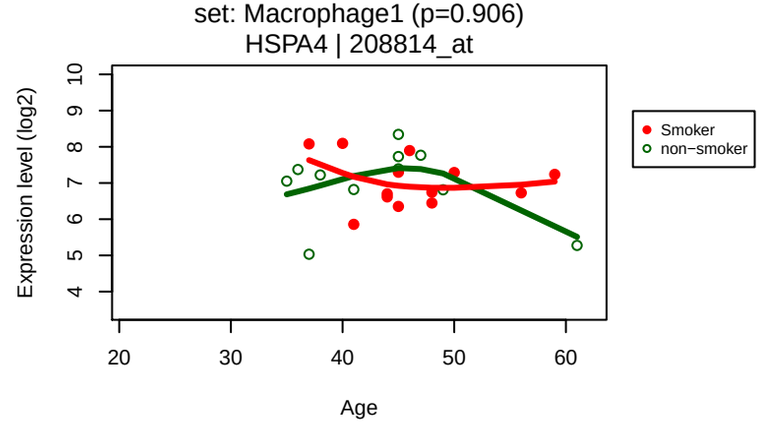

set: SmallAirway1 (p=0.2835)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker

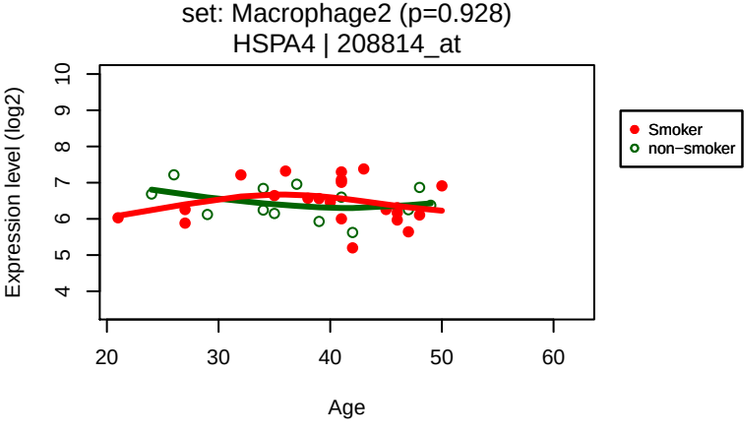

set: SmallAirway2 (p=0.6198)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker

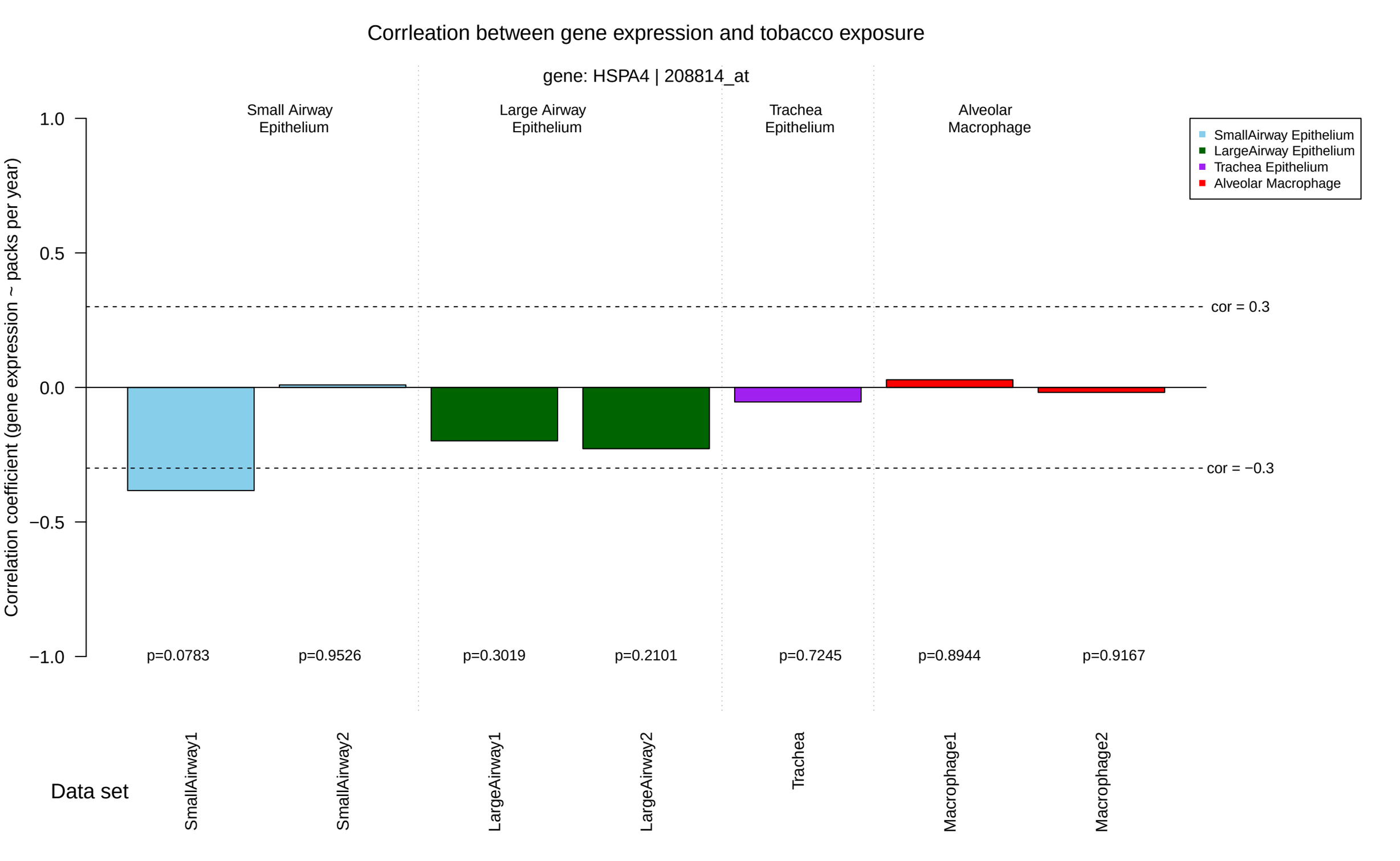

set: LargeAirway1 (p=0.0538)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9427)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8013)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.906)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.928)
HSPA4 | 208814_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA4 | 208814_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0783
p=0.9526
p=0.3019
p=0.2101
p=0.7245
p=0.8944
p=0.9167
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2